Total No. of Questions : 8]

SEAT No. :

PC2485

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[6354]-614

B.E. (Mechanical)

TURBOMACHINERY

(2019 Pattern) (Semester- VII) (402043)

Time : 2 Hours] ***[Max. Marks : 50***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagram must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data wherever necessary.***
5) ***Use of steam table/Mollier chart is allowed.***

Q1) a) Give classification of turbomachines with suitable example. **[6]**

b) A Pelton wheel is having a mean bucket diameter of 1 m and is running at 1000 rpm. The net head on pelton wheel is 700m. If the side clearance angle is 15° and discharge through the nozzle is 0.1 m^3/s. Determine **[8]**

i) Power available at the nozzle.

ii) Hydraulic efficiency of the turbine.

OR

Q2) a) Explain construction and working of Kaplan Turbine with neat sketch. **[6]**

b) The external and internal diameters of inward flow reaction turbines are 1.20 m and 0.6 m respectively. The head on the turbine is 22 m and velocity of flow through the runner is constant and equal to 2.5 m/s. The guide blade is given as 10° and runner vanes are radial at inlet. If the discharge at outlet is radial, determine **[8]**

i) The speed of the turbine

ii) The vane angle at the outlet of the runner

iii) Hydraulic efficiency of the turbine.

P.T.O.

Q3) a) What is compounding? Explain the need of compounding. Explain any one method of compounding in steam turbines. **[6]**

b) In De Laval turbine, steam is issued from the nozzle with a velocity of 1500 m/s whereas the mean blade velocity is 500m/s. The nozzle angle is 20° and the inlet and outlet angles of blades are equal. The mass of steam flowing through the turbine is at the rate of 1200kg/hr. Assuming blade velocity coefficient k=0.8, Draw velocity diagram and determine **[6]**

i) The Blade angles

ii) The power developed by turbine

iii) The Blade Efficiency

OR

Q4) a) Explain governing of steam turbine with any one method. **[6]**

b) In Parson's reaction turbine running at 500rpm with 50% reaction develops 75 kW per kg per second of steam. The exit angle of blades is 20º and steam velocity is 1.5 times the blade velocity. Determine **[6]**

i) Blade Velocity

ii) Inlet angle of moving blade.

Q5) a) State & explain: **[6]**

i) Unit Speed

ii) Unit Discharge

iii) Unit Power

b) The outer diameter of an impeller of a centrifugal pump is 400mm and outlet width is 50mm. The pump is running at 800 rpm and is working against a total head of 15 m. The vane angle at outlet is 40° and manometric efficiency is 75% Determine **[6]**

i) Velocity of flow at outlet

iii) Velocity of water leaving the vane

iii) Angle made by absolute velocity at outlet

OR

Q6) a) Explain various heads in centrifugal pump with a neat sketch. **[6]**

b) A centrifugal pump delivers 1565 LPS against a manometric head of 6.1m. When the impeller rotates at 200 rpm. The impeller diameter is 1.22 m and area at outer periphery is 6450 cm^2. If the vanes are set back at angle of 26° at the outlet , Determine **[6]**

i) Manometric Efficiency

ii) Power required to drive the pump

iii) Minimum starting speed if ratio of external to internal is 2.

Q7) a) Explain Construction and working of axial flow compressor with a neat sketch. **[4]**

b) A rotary air compressor working between 1 bar and 2.5 bar has internal and external diameter of impeller as 300 mm and 600 mm respectively. The vane angle at inlet and outlet are 30° and 45° respectively. If air enters impeller at 15 m/s, Find **[8]**

i) Speed of impeller in rpm

ii) Work done by compressor per kg of air.

OR

Q8) a) Differentiate between centrifugal compressor and axial flow compressor. **[4]**

b) The impeller of the centrifugal compressor has the inlet and outlet diameter of 0.3 and 0.6 m respectively. The intake is from the atmosphere at 100 kPa and 300 K, without any whirl component. The outlet blade angle is 75°. The speed is 10000 rpm and velocity of flow is constant at 120 m/s. If the blade width at inlet is 6 cm, determine the following **[8]**

i) Specific work

ii) Exit pressure

iii) Mass flow rate

iv) Power required to compressor if the overall efficiency is assumed to be 0.7.